Ethical Hacking and Advanced Cybersecurity (DSE-3/GE-5)

CREDIT DISTRIBUTION, ELIGIBILITY AND PRE-REQUISITES OF THE COURSE

Course title	Credits	Credit distribution of the course			Eligibility criteria	Prerequisite of the course (if any)
		Lecture	Tutorial	Practical		
Ethical Hacking and Advanced Cybersecurity	4	3L	0T	1P	-	Fundamentals of Cybersecurity

Course Hours: L: 03 T: 00 P: 02

Course Outcomes

At the end of this course, students will be able to:

1. Understand and apply the principles of ethical hacking and penetration testing.
2. Detect and exploit vulnerabilities in a controlled and responsible manner.
3. Utilize advanced tools and frameworks for ethical hacking and network defense.
4. Conduct forensic investigations and malware analysis to identify the root causes of breaches.
5. Formulate and implement advanced cybersecurity strategies to protect critical infrastructure.

Course Objectives

1. Introduce ethical hacking techniques and practices to identify vulnerabilities in systems and networks.
2. Provide insights into advanced cybersecurity measures, including malware analysis and threat hunting.
3. Equip students with tools for ethical hacking, such as Kali Linux, Metasploit, and Burp Suite.
4. Develop ethical responsibility and compliance with legal frameworks in cybersecurity.
5. Enable hands-on experience in simulating and defending against sophisticated cyber-attacks.

Unit 1: Fundamentals of Ethical Hacking

Definition, Scope, and Importance of Ethical Hacking; Understanding Cyber Laws and Ethical Hacking Ethics; Types of Hackers (White Hat, Black Hat, Grey Hat); Phases of Ethical Hacking (Reconnaissance, Scanning, Gaining Access, Maintaining Access, Covering Tracks); Introduction to Ethical Hacking Tools such as Kali Linux, Metasploit, and Wireshark.

Unit 2: Network and System Security

Overview of Network Security: Firewalls, IDS/IPS, VPNs, Proxy Servers; Vulnerability Assessment and Penetration Testing (VAPT) Techniques and Tools; Network Hacking: Packet Sniffing, Spoofing, and Man-in-the-Middle (MITM) Attacks; System Security Concepts: Privilege Escalation, Backdoors, and Trojans; Introduction to Wireless Network Security and WPA/WPA2 Cracking.

Unit 3: Application and Web Security

Understanding Web Application Attacks: SQL Injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF); Securing Web Applications: OWASP Top 10 Framework; Exploring Secure Coding Practices; Hacking APIs and Web Services; Introduction to Mobile Application Security.

Unit 4: Advanced Cybersecurity Concepts

Incident Detection and Response: Monitoring and Analyzing Cyber Attacks; Cryptographic Techniques: Encryption Standards, Digital Signatures, and Certificates; Advanced Persistent Threats (APTs): Detection and Mitigation

Strategies; Exploring Blockchain Security and Zero-Trust Architecture; Overview of Emerging Threats in AI and IoT.

Practical Component

1. Conducting reconnaissance using tools like Maltego to gather intelligence on a target system.
2. Exploiting vulnerabilities in a controlled lab environment to understand the ethical hacking lifecycle.
3. Developing exploits using custom scripts and testing them in a sandboxed environment.
4. Performing secure code analysis to identify common vulnerabilities like SQL injection or XSS.
5. Building and deploying honeypots to monitor and log potential attack attempts.
6. Simulating ransomware attacks and crafting effective incident response strategies.
7. Conducting security audits of cloud platforms to ensure compliance with security standards.

Suggested Readings

1. "The Hacker Playbook 3: Practical Guide To Penetration Testing" by Peter Kim
2. "Hacking: The Art of Exploitation" by Jon Erickson
3. "Penetration Testing: A Hands-On Introduction to Hacking" by Georgia Weidman
4. Hacking: The Art of Exploitation by Jon Erickson (No Starch Press)
5. Penetration Testing: A Hands-On Introduction to Hacking by Georgia Weidman (No Starch Press)
6. The Web Application Hacker's Handbook by Dafydd Stuttard and Marcus Pinto (Wiley)
7. Metasploit: The Penetration Tester’s Guide by David Kennedy, Jim O’Gorman, Devon Kearns, and Mati Aharoni (No Starch Press)

List of Experiments

Note: The course instructor will design experiments/mini-projects to complete the practical component of the course.